

EEE

2023

(1) a) The resistance R of a material depends mainly on these factors
1. Resistivity (ρ) - depends on the type of material and its temperature:

2. Length (L) - the longer the material the longer the resistance.

3. Area (A) - the thicker (wider) the material, the less resistance.

These factors relate through the formula:-

$$R = \rho \times \frac{L}{A}$$

where, R = resistance, ρ = resistivity, L = length, A = Area.

b) Ammeter is connected in series because.

- It measures the current flowing through the circuit.
- To measure current accurately, it must be part of the current path.
- It has very low resistance to avoid changing the actual current.

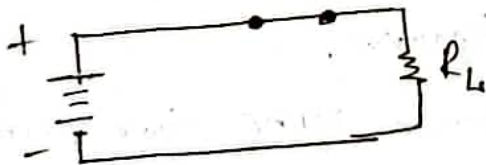
Voltmeter is connected in parallel because:

- It measures the potential difference across two points.
- To correctly measure voltage, it must be connected across the component.
- It has a very high resistance to prevent drawing extra current from the circuit.

(2) (b) In a parallel connection, the total resistance R_{total} is found by, $\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$

- In a parallel connection, each new resistor provides an additional path for the current to flow.
 - More path means less overall resistance to the flow of current.
 - Even if the new resistor has a high resistance, it still adds a new path and reduces the total resistance.
- In short, Adding any resistance in parallel always reduces the total resistance, no matter how large or small the added resistor is.

(3) (b) Short Circuit: A circuit where current flows through an unintended path, typically with very low resistance, which can cause excessive current flow and potentially damage components or cause fire hazards.



Open Circuit: A circuit where the current cannot flow because there is a break or gap in the path, like when a switch is off or a wire is disconnected.



(C) (b) For N capacitors $C_1, C_2, C_3, \dots, C_N$ connected in series, the equivalent capacitance C_{eq} is given by:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$$

$$\text{or, } C_{eq} = \frac{1}{\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}\right)}$$

When n capacitors are connected in series the total capacitance is given by the reciprocal of the sum of the reciprocals of the individual capacitance, this is because-

1. In series, the total capacitance is always less than the smallest individual capacitance.

2. The equivalent capacitance decreases as more capacitors are added in series.

(*) (a) Faraday's law of electromagnetic induction states that, the induced electromotive force (EMF) in any closed circuit is equal to the rate of change of magnetic flux through the circuit.

$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

where, $\mathcal{E}$ = induced EMF

Φ_B = ~~magnet~~ magnetic flux;

Here, Φ_B is a magnetic flux, given by the product of the magnetic field B and the area A through which

the field passes, $\Phi_B = B \cdot A \cos \theta$

The negative sign in the equation is explained by Lenz's Law states that the direction of the induced current.



$$\text{Voltage, } V = V_0 \sin \theta$$

$$\text{Current, } I = I_0 \sin \left(\theta - \frac{\pi}{2} \right)$$

We know in an inductive circuit current lags voltage by $\frac{\pi}{2}$ angle.

$$\therefore \text{Power } P = VI$$

$$= V_0 \sin \theta \cdot I_0 \sin \left(\theta - \frac{\pi}{2} \right)$$

$$= V_0 I_0 \sin \theta \cdot (-\cos \theta)$$

$$= -\frac{V_0 I_0}{2} (2 \sin \theta \cos \theta)$$

$$= -\frac{V_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} \cdot \sin 2\theta$$

$$= -V_{\text{rms}} \cdot I_{\text{rms}} \sin 2\theta$$

For pure inductive circuit of full cycle,

$$\theta = \frac{\pi}{2}$$

$$\therefore P = -V_{\text{rms}} \cdot I_{\text{rms}} \cdot \sin \left(2 \times \frac{\pi}{2} \right)$$

$$\therefore P = 0$$

So, the average power or power dissipated by the ideal inductor is zero watts.

② ① Admittance:

Admittance is the measure of how easily a circuit allows current to flow when a voltage is applied. It is the reciprocal of impedance.

$$Y = \frac{1}{Z}$$

Susceptance:

Susceptance is the imaginary part of admittance and represents the circuit's ability to conduct reactive current.

$$B = \frac{1}{X}$$

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③ ② Supernode: In a circuit analysis, a supernode occurs when two or more nodes are connected by a voltage source. (either independent or dependent).

Ex: If nodes A & B are connected by a 5V voltage source, you treat A and B together as one supernode.

$$V_A - V_B = 5V$$

Supermesh: In a circuit analysis, a supermesh occurs when two meshes are connected by a ~~current~~ current source. (either independent or dependent).

Ex: If two loops share a current source, combine the two loops into one supermesh.

$$I_1 - I_2 = \text{Value of current source.}$$

⑤ ⑥ condition for maximum power:-

The load resistance R_L must be equal to the thevenin resistance R_{th} of the source network seen from the load terminals. $R_L = R_{th}$.

⑥ ⑥ The voltage across an inductor is,

$$V = L \frac{di}{dt}$$

Power delivered to the inductor,

$$P = V \times i$$

$$P = L \frac{di}{dt} \times i$$

$$\Rightarrow P(t) = Li \frac{di}{dt}$$

- the energy stored is,

$$W = \int_{-\infty}^t P(t) dt$$

$$= L \int_{-\infty}^t \frac{di}{dt} i dt$$

$$= L \int_{-\infty}^t i di$$

$$= \frac{1}{2} Li^2(t) - \frac{1}{2} Li^2(-\infty) + c$$

$$i(-\infty) = 0$$

$$W = \frac{1}{2} Li^2$$

$\therefore$ Energy stored in an inductor $W = \frac{1}{2} Li^2$

⑦ @ From Ohm's Law,

$$v(t) = i(t) R$$

$$\begin{aligned}\therefore P(t) &= v(t) \cdot i(t) \\ &= i(t) \cdot R \cdot i(t) = R \cdot i^2(t)\end{aligned}$$

$$\begin{aligned}P_{avg} &= \frac{1}{T} \int_0^T P(t) dt \\ &= \frac{1}{T} \int_0^T R i^2(t) dt \\ &= \frac{R}{T} \int_0^T i^2(t) dt\end{aligned}$$

Assume the current is sinusoidal:

$$i(t) = I_0 \sin(\omega t)$$

$$i^2(t) = I_0^2 \sin^2(\omega t)$$

$$\therefore P_{avg} = R \cdot \frac{1}{T} \int_0^T (I_0 \sin(\omega t))^2 dt$$

$$P_{avg} = R \cdot I_0^2 \cdot \frac{1}{2}$$

$$P_{avg} = \frac{1}{2} R I_0^2$$

$$\text{We know, } I_{rms} = \frac{I_0}{\sqrt{2}}$$

$\therefore$ The average power is

$$P_{avg} = R I_{rms}^2$$

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⊗ (c) Instantaneous Power:-

Instantaneous power is the power at any given moment in time.

$$P(t) = v(t) \cdot i(t)$$

* Average Power:-

Average power is the total energy consumed by a circuit over a complete cycle, divided by the time period of the cycle.

$$P_{avg} = R I_{rms}^2$$

* Effective Power or RMS value, is a measure of the equivalent DC value that would produce the same amount of heating ^{or power dissipation} in a resistor as the AC signal.

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} \quad \text{or} \quad V_{rms} = \frac{V_{max}}{\sqrt{2}}$$

* Complex power is a quantity used to describe the power in AC circuits that involve both real power and reactive power.

$$S = P + jQ$$

$\downarrow$ $\downarrow$
real reactive
power power

④ (a) For a capacitor,
 $Q = CV$

$$p(t) = v(t) \cdot i(t)$$

$$p(t) = v \cdot \frac{dQ}{dt}$$

$$p(t) = v \cdot \frac{d(CV)}{dt}$$

$$p(t) = v \cdot C \cdot \frac{dv}{dt}$$

$$W = \int_0^T p(t) dt$$

$$W = \int_0^T v C \frac{dv}{dt} dt$$

$$W = C \int_0^v v dv$$

$$W = C \left[\frac{v^2}{2} \right]_0^v$$

$$W = \frac{1}{2} CV^2$$

⑥ (a) The substitution theorem states that, in a linear electrical network, a branch can be replaced by an equivalent branch, and the overall behaviour of the network will remain unchanged.

⑦ (a) Limitations:

1. Valid for Linear Networks.
2. Cannot handle Non linear Elements.
3. Only for single source networks.
4. AC Circuits must be Bilateral.
5. Does not apply to time-varying systems.

* **Reciprocity theorem** states that, the ratios of voltage to current remains the same when the positions of the source and the response are interchanged.

* **Resistance** is the opposition to the flow of electric current in a material.

* **Current** is the time rate of change of charge, measured in amperes (A).

$$i = \frac{dq}{dt}$$

* **Voltage** is the energy required to move a unit charge from a reference point (-) to another point (+), measured in volts (V).

$$v = \frac{dw}{dq}$$

* **Power** is the time rate of expending or absorbing energy, measured in watts (W).

* **Energy** is the capacity to do work measured in joules (J).

* A **capacitor** is an electrical component that stores energy in the form of an electric field.

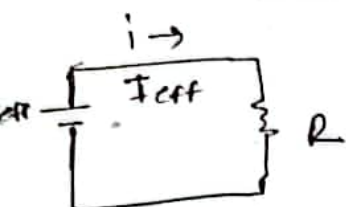
* A **inductor** is an electrical component that stores energy in the form of magnetic field.

Derive average power :-



Average power,

$$P = \frac{1}{T} \int_0^T i^2 R dt \quad \text{--- (i)}$$



$$P = I_{\text{eff}}^2 R \quad \text{--- (ii)}$$

@ circuit,

$$I_{\text{eff}}^2 R = \frac{1}{T} \int_0^T i^2 R dt$$

$$I_{\text{eff}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$i = I_m \sin \omega t$$

$$I_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T (I_m \sin \omega t)^2 dt}$$

$$= \sqrt{\frac{1}{T} \int_0^T I_m^2 \sin^2 \omega t dt}$$

$$= \sqrt{\frac{I_m^2}{2T} \int_0^T 2 \sin^2 \omega t dt}$$

$$= \sqrt{\frac{I_m^2}{2T} \int_0^T (1 - \cos 2\omega t) dt}$$

$$= \sqrt{\frac{I_m^2}{2T} \left[\int_0^T dt - \int_0^T \cos 2\omega t dt \right]}$$

$$= \sqrt{\frac{I_m^2}{2T} \cdot [t]_0^T - \left[\frac{\sin 2\omega t}{2\omega} \right]_0^T}$$

$$= \sqrt{\frac{I_m^2}{2T} [(T-0) - \frac{\sin 2\omega t - \sin 2\omega \cdot 0}{2\omega}]}$$

$$= \sqrt{\frac{I_m^2}{2T} \times T}$$

$$= \frac{I_m}{\sqrt{2}}$$

$$\therefore P_{\text{avg}} =$$

$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$